

GROUP NUMBER: 35

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TEST: The goal is to test the performance of **Sticky Sampling (SS)** and **Count-Min (CM)** for different input configurations. We refer to an item as *frequent/almost frequent/rare* if its true frequency is in the interval $[n\phi, +\infty) / [n(\phi - \epsilon), n\phi)$, $[0, n(\phi - \epsilon))$, respectively.

You must fill in the following tables using the input configurations indicated in red for each table. You must run each test three times and report the average values of the three runs

STICKY SAMPLING: $n=1000000$, $\phi=0.07$, $\delta=0.05$, port=8888				
ϵ	Number of frequent items returned by SS	Number of almost frequent items returned by SS	Number of rare items returned by SS	Number of elements stored in the dictionary used by SS
0.01	10	0	0	36
0.02	10	0	0	23
0.04	10	3	0	19

s

COUNT MIN: $n=1000000$, $\phi=0.07$, $d=5$, port=8888				
w	Number of frequent items returned by CM	Number of almost frequent items (w.r.t. $\epsilon=0.04$) returned by CM	Number of rare items (w.r.t. $\epsilon=0.04$) returned by CM	Total number of items returned by CM (sum of previous 3 columns)
15	10	0	214	224
30	10	0	14	24
60	10	0	2	12

ANSWER THE FOLLOWING QUESTIONS

- Did you use some Generative AI tool? Yes
- (Only if you answered YES to previous question) Describe in at most 10 lines how you used the Generative AI tool.

I used Claude AI to verify parts of my code, such as the Count-Min Sketch logic. I also used AI to automatically generate the true/almost/rare frequencies instead of calculating them by hand, printing them directly to the terminal. After reviewing these AI's code parts, the results appear to be correct.